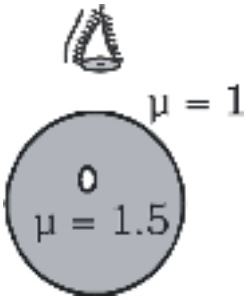
PART 1 : PHYSICS

SECTION-I (i)

1) There is a small air bubble inside a glass sphere (μ = 1.5) of radius 10 cm. The bubble is 4 cm below the surface and is viewed normally from the outside as shown in figure. Find the apparent depth of the bubble.



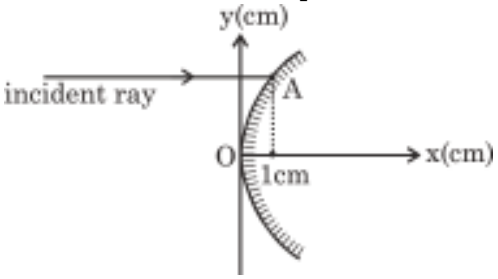
(A) 10 cm

(B) 3.1 cm

(C) 5.3 cm

(D) 6.2 cm

2) A light ray is incident parallel to x-axis, at point A of a convex mirror $x = \frac{4}{9}y^2$, where x & y are in cm. If x-coordinate of point A is x = 1cm then angle made by reflected ray with +ve x-axis is :



(A) 37°

(B) 53°

(C) 74°

(D) 90°

3) The roadway of a bridge over a canal is in the form of a circular arc of radius a. The greatest speed with which a motor cyclist can cross the bridge without leaving the roadway at the highest point in its path

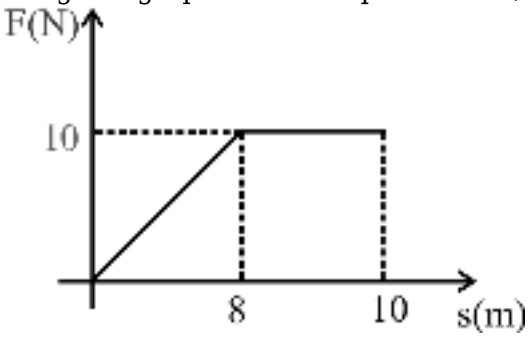
(A) $\sqrt{2ag}$

(B) $3\sqrt{ag}$

(C) $\sqrt{ag}$

(D) $2\sqrt{ag}$

4) A force is applied on the body of mass 2 kg which varies with displacement of body according to the given graph. If initial speed is 2 m/s, find the final speed of the body :-



(A) 8 m/s

(B) 6 m/s

(C) 4 m/s

(D) 10 m/s

SECTION-I (ii)

1) A beam of light parallel to main optical axis falls on a converging lens of focal length f_1 = 20 cm. Behind the lens, coaxially at some distance L, is diverging lens of focal length f_2 = – 20 cm. After passing through the diverging lens, the light focuses at a point 'A' 5 cm behind the lens. The position of lens is then interchanged :-

(A) The value of L is 16 cm.

(B) The value of L is 80/3 cm.

(C) Point A shifts by 40 cm after the lenses are interchanged

(D) Point A shifts by 125/3 cm after the lenses are interchanged

2) A man pushes against a rigid, immovable wall while standing on rough ground. Which of the following statement(s) is/are wrong concerning this situation ?

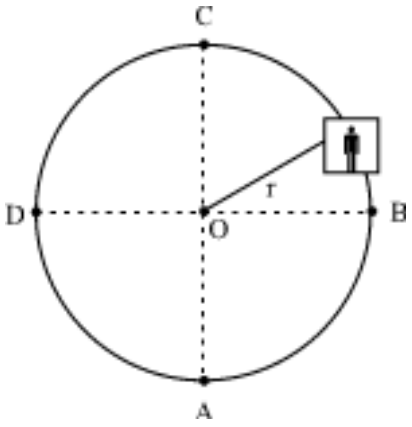
(A) The man can never exert a force on the wall which exceeds his weight

(B) If the man pushes on the wall with a force of 200N, we can be sure that the wall is pushing back with a force of exactly 200 N on him.

(C) Since the wall cannot move, it cannot exert any force on the man

(D) The man cannot be in equilibrium, since he is exerting a net force on the wall.

3) A machine, in an amusement park, consists of a cage at the end of one arm, hinged at O. The cage revolves along a vertical circle of radius r (A B C D) about its hinge O, constant linear speed $v = \sqrt{gr}$. The cage is so attached that the man of weight 'w' standing on a weighing machine, inside the



cage, is always vertical. Then:

(A) The weight reading at A is greater than the weight reading at C by 2w

(B) The weight reading at D = w

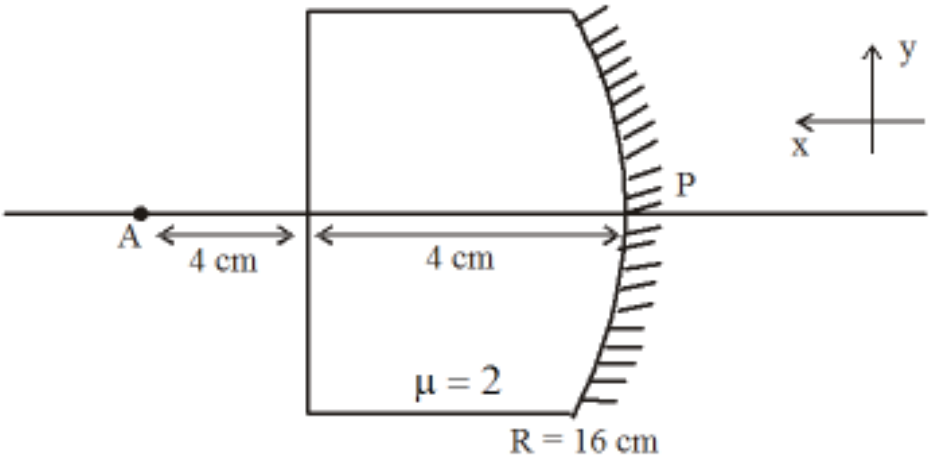
(C) The ratio of the weight reading at C to that at A = 0

(D) The ratio of the weight reading at A to that at B = 2

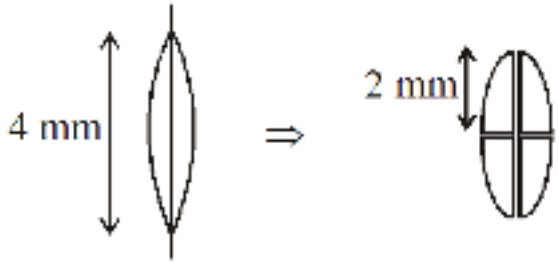
SECTION-II

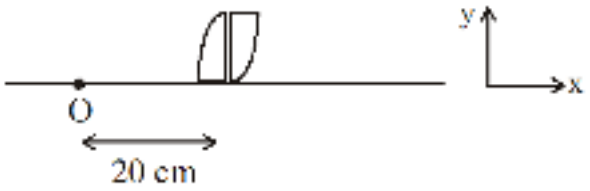
1) The speed of light in glass of refractive index 1.5 is $2 \times 10^8 m/s$. In a certain liquid the speed is $2.5 \times 10^8 m/s$. The refractive index of the liquid is :-

2) A curved thick glass surface is silvered at curved face & not silvered on plane surface. Object is placed at A as shown in figure. Considering P (pole of the silvered surface) as origin. If x - co-ordinate of final image is 'n' cm then find n.

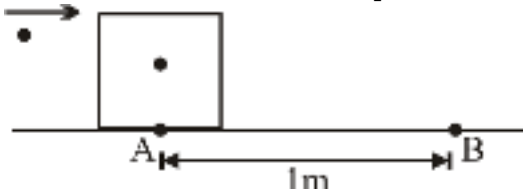


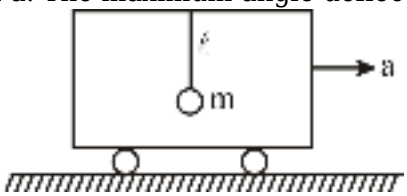
3) An equiconvex lens having μ = 1.5 and radius of curvature 10 cm is cut into four equal parts as shown in the diagram. Now two parts are used to form



the image as shown in the diagram 

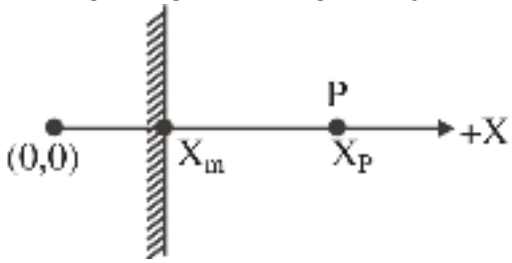
Now consider O as origin (0, 0) if x and y co-ordinate of image is (n cm, m cm) then value of $\frac{n}{m}$ is.....

4) A bullet of mass 0.01 kg strikes a block of mass 0.09 kg and gets embedded in it. The block then slide along the surface and comes to rest at point B. The coefficient of friction varies as $\mu(x)=0.1x$ along the surface where AB = 1 m and x is measured from point A. Find the velocity (in m/s) with which the bullet hits the block. 

5) In shown figure, the trolley starts accelerating with acceleration a. The maximum angle deflected by thread from vertical will be $y \tan^{-1}\left(\frac{xa}{g}\right)$ then the value of $\frac{x}{y}$ is ? 

6) An escalator is used to move 5 people (60 kg each) per minute from the first floor of a department store to the second floor, 20 m above. Neglecting friction calculate the power (in kW) required.

SECTION-I (iii)

1) The position of a mirror and a particle moving along X-axis is given by $x_m = t^3 - 6t^2$ and $x_p = 9t^2 - 60t$. Match the column-I with column-II. 

Column-I		**Column-II**	
(A)	Velocity of image of particle is zero	(P)	2 sec
(B)	Acceleration of image of particle is zero	(Q)	3 sec
(C)	Acceleration of image of particle is negative	(R)	3.5 sec
(D)	Velocity of image of particle is towards the origin	(S)	4 sec
		(T)	5 sec

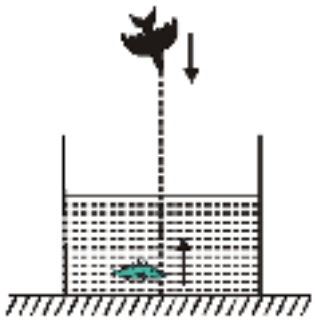
(A) A → PT;B → R;C → PQ;D → QRS

(B) A → P;B → Q;C → R;D → S

(C) A → PQ;B → R;C → S;D → T

(D) A → P;B → Q;C → R;D → T

2) A bird in air is diving vertically over a tank with speed 6 cm/s. Base of the tank is silvered. A fish in the tank is rising upward along the same line with speed 4 cm/s. [Take: $\mu_{water} = 4/3$]



List-I		List-II	
(A)	Speed (in cm/s) of the image of fish as seen by the bird directly.	(P)	12
(B)	Speed (in cm/s) of the image of fish formed after reflection from the mirror as seen by the bird.	(Q)	4
(C)	Speed (in cm/s) of image of bird relative to the fish looking upwards.	(R)	9
(D)	Speed (in cm/s) of image of bird relative to the fish looking downwards in the mirror	(S)	3
		(T)	20/3
		(U)	32/3

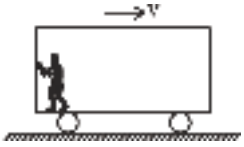
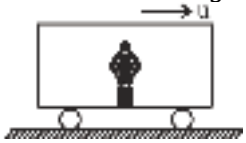
(A) A → P;B → Q;C → R;D → S

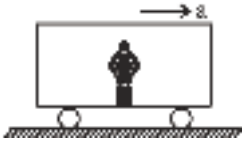
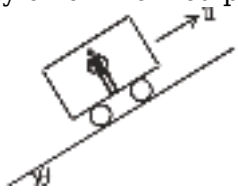
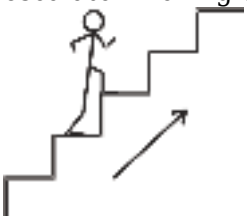
(B) A → P;B → R;C → S;D → T

(C) A → R;B → S;C → P;D → Q

(D) A → P;B → S;C → R;D → T

3) Match the column

Column-I All observation are seen from ground		Column-II	
(A)	Net work done by normal force exerted by all surfaces on man ≠ 0	(P)	Man pushing the wall and the bus is moving with constant velocity 
(B)	Net work done by normal force exerted by all surfaces on man = 0	(Q)	Man standing in bus and bus is moving with constant velocity 

(C)	Work done by friction force on man = 0	(R)	Man standing in bus and bus is accelerated with constant acceleration a 
(D)	Work done by friction force on man ≠ 0	(S)	Man standing in bus and bus is moving with constant velocity on an inclined plane having inclination θ. 
		(T)	A man is walking up ward uniformly on the escalator moving up with uniform velocity 

(A) A → PQ;B → RS;C → S;D → T

(B) A → P;B → Q;C → R;D → S

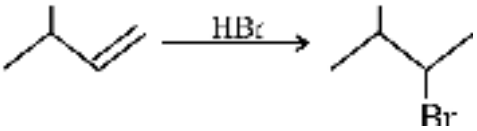
(C) A → PT;B → QR;C → ST;D → PRS

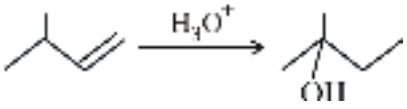
(D) A → PT;B → QRS;C → QT;D → PRS

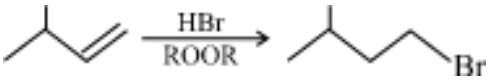
PART 2 : CHEMISTRY

SECTION-I (i)

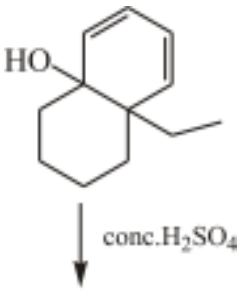
1) Reaction incorrectly matched with its major product is :

(A) 

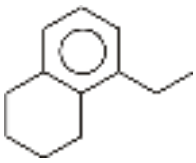
(B) 

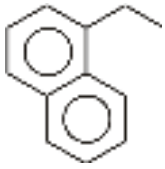
(C) 

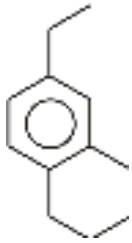
(D) None of these

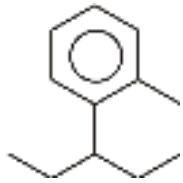


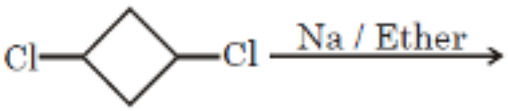
2) The major product in the following reaction

(A) 

(B) 

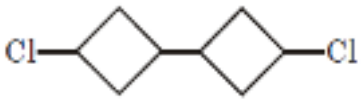
(C) 

(D) 



3)

Major product of above reaction :

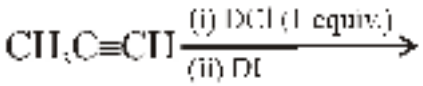
(A) 

(B) 

(C) 

(D) 

4) The major product of the following reaction is :



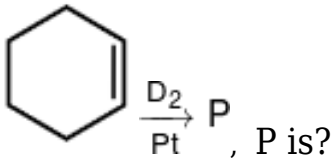
(A) $CH_3CD(Cl)CHD(I)$

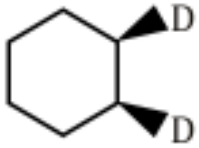
(B) $CH_3CD_2CH(Cl)(I)$

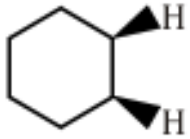
(C) $CH_3CD(I)CHD(Cl)$

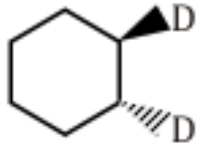
(D) $CH_3C(I)(Cl)CHD_2$

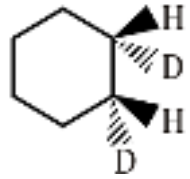
SECTION-I (ii)

1)  , P is?

(A) 

(B) 

(C) 

(D) 

2) The ionic addition of HCl to which of the following compounds will produces a compound having Cl on carbon next to terminal.

(A) $CF_3.(CH_2)_3.CH=CH_2$

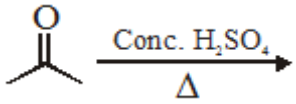
(B) $CH_3.CH=CH_2$

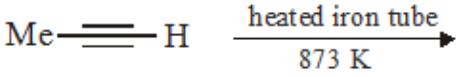
(C) $CF_3.CH=CH_2$

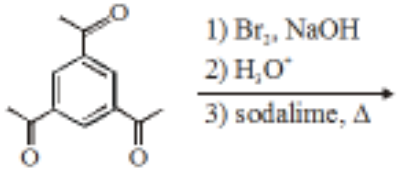
(D) $CH_3.CH_2CH=CH.CH_3$

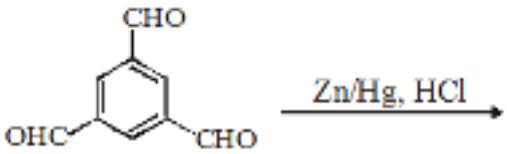
3)

The reaction(s) leading to the formation of 1,3,5-trimethylbenzene is (are)

(A) 

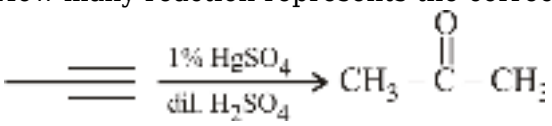
(B) 

(C) 

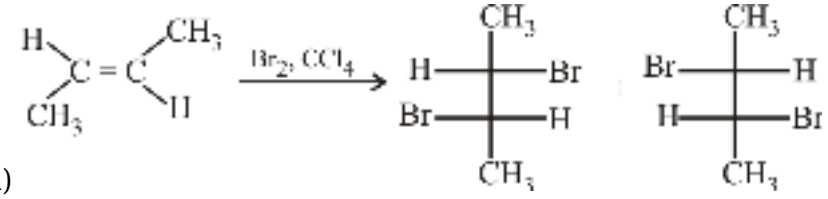
(D) 

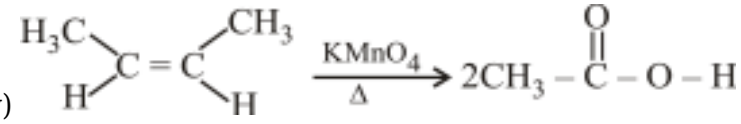
SECTION-II

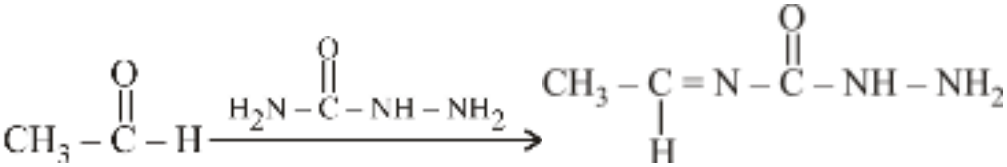
1) How many reaction represents the correct major product

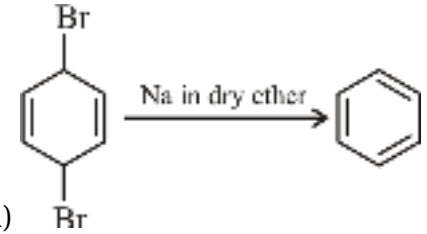
(i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) 

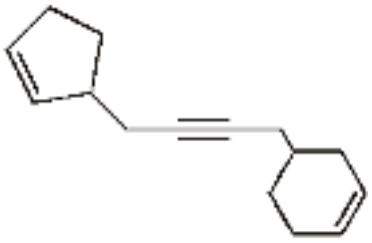
(vii) $CH_3-C\equiv C-CH_3 \xrightarrow{H_2,\ Pd\text{–}BaSO_4} CH_3-CH_2-CH_2-CH_3$

(viii) $CH_3-CH=CH_2 \xrightarrow{HCl,\ R\text{–}O\text{–}O\text{–}R} CH_3-CH_2-CH_2Cl$

$NO_2-CH_2-CH_2Br$

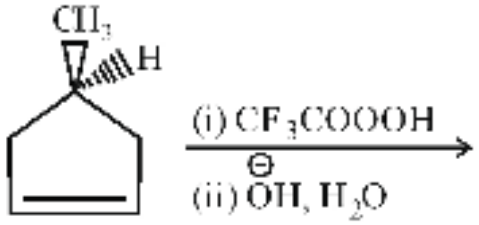
(ix) $NO_2-CH=CH_2 \xrightarrow{HBr}$

2) Total number of hydroxyl groups present in a molecule of the major product P is ___

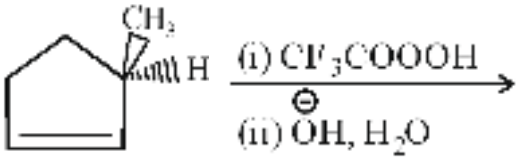


P

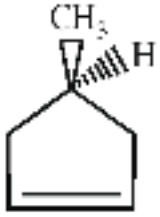
3) The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is _____.

4) In the given reactions:- (i) 

X + Y

(ii) 

P + Q

(iii) 

A + B

(iv) 

M + N

Number of correct statement(s) among the following are :-
(1) X and Y are enantiomers
(2) M and N are enantiomers
(3) A and B are diastereomers
(4) (X and Y) are identical with (A and B)
(5) (P and Q) are identical with (M and N)
(6) M and N are geometrical isomers

5) How many alkene on catalytic reduction gives following product.
Alkene $\xrightarrow{H_2/Pt}$ Iso-pentane

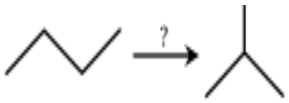
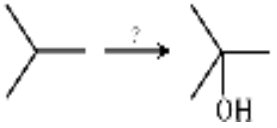
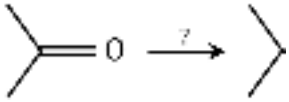
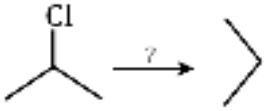
6) (i) Cis - 2-Butene $\xrightarrow[CCl_4]{Br_2}$ X

(ii) Trans-cyclo octene $\xrightarrow[CCl_4]{Br_2}$ Y

(a) Number of organic products obtained in x
(b) Number of organic products obtained in y
The VALUE OF x + y

SECTION-I (iii)

1) **Match the column:**

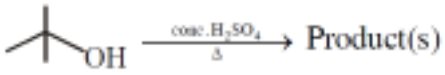
	Column-I		**Column-II**
a.		p	Alkaline $KMnO_4$
b.		q	HI + Red P
c.		r	$AlCl_3$ + HCl at 298 K
d.		s	NH_2–NH_2 + $EtO^{\ominus}$
		u.	Zn – Hg/HCl

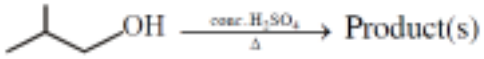
(A) a → r;b → p;c → q,s,u;d → q
(B) a → q,s,u;b → p;c → r;d → q
(C) a → r;b → r;c → q,s,u;d → q
(D) a → q;b → p;c → q,s,u;d → r

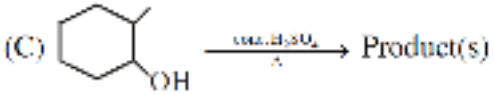
2)

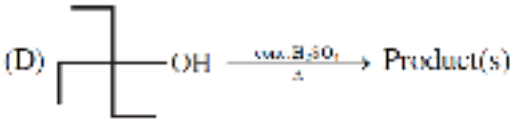
Match the following reaction with their corresponding pattern of graph :

Column-I
(Conversions)

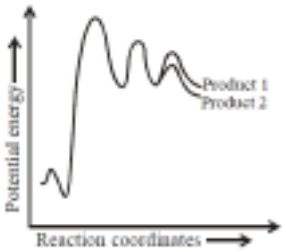
(A) 

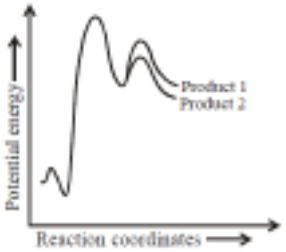
(B) 

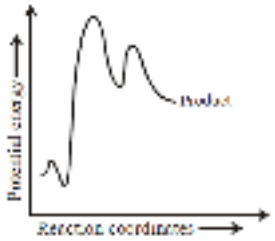
(C) 

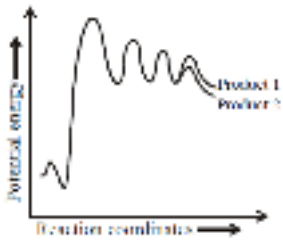
(D) 

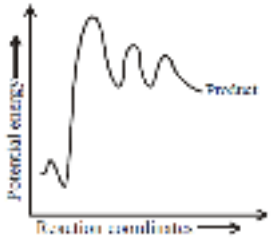
Column-II
(Reagents)

(P) 

(Q) 

(R) 

(S) 

(T) 

(A) A → P;B → T;C → R;D → R

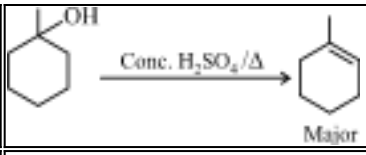
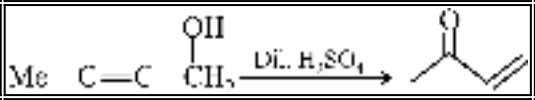
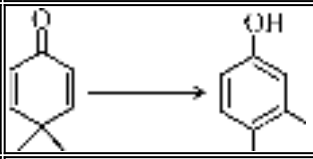
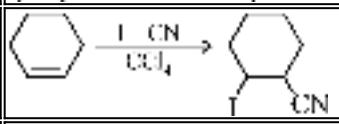
(B) A → Q;B → S;C → P;D → R

(C) A → R;B → T;C → P;D → R

(D) A → R;B → S;C → P;D → T

3)

Column - I (Chemical reaction)	**Column - II (Type of reaction/intermediate involve)**

(P)		(1)	Elimination reaction
(Q)		(2)	Addition reaction
(R)		(3)	Carbocation intermediate is formed [Classical /Non classical]
(S)		(4)	Rearrangment reaction
		(5)	Racemic mixture is formed

(A) P → 1;Q → 3;R → 3;S → 3

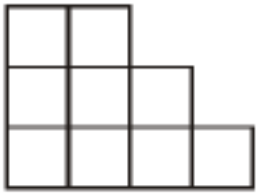
(B) P → 1;Q → 2;R → 3;S → 3

(C) P → 1;Q → 2;R → 3;S → 4

(D) P → 1;Q → 3;R → 3;S → 4

PART 3 : MATHEMATICS

SECTION-I (i)

1) The number of ways in which 5 X's can be placed in the squares of the figure so that no row remains empty is 

(A) 97

(B) 98

(C) 100

(D) 126

2) A grid is formed by taking x, y ∈ I such that $|x| \le 5$, $|y - 1| \le 6$ and $xy \ge 0$, then number of ways of reach from (–5, –5) to (5, 7) by using shortest path is

(A) $\frac{\lfloor 10}{\lfloor 5 \lfloor 5} \times \frac{\lfloor 12}{\lfloor 7 \lfloor 5}$

(B) $\frac{\lfloor 22}{\lfloor 10 \lfloor 12}$

(C) $\frac{\lfloor 9}{\lfloor 4 \lfloor 4} \times \frac{\lfloor 11}{\lfloor 6 \lfloor 4}$

(D) $\frac{\lfloor 20}{\lfloor 9 \lfloor 11}$

3) Value of the determinant $\begin{vmatrix} \sin 2010\pi & \sin\frac{\pi}{5}\sin\frac{2\pi}{5} & \tan\frac{667\pi}{4} \\ \sin\frac{3\pi}{5} & \cos 2009\frac{\pi}{2} & \cos 2013\frac{\pi}{6} \\ \tan\frac{3\pi}{10} & \cot 2011\frac{\pi}{2} & \sin\frac{4\pi}{5} \end{vmatrix}$ is

(A) $\frac{5}{16}$

(B) $\frac{5}{8}$

(C) $\frac{-5}{16}$

(D) 1

4) If the system of linear equations, x + 2ay + az = 0, x + 3by + bz = 0 and x + 4cy + cz = 0 has a non-zero solution then $b\left(\frac{1}{a}+\frac{1}{c}\right)$

(A) 1

(B) 2

(C) 3

(D) 4

SECTION-I (ii)

1)

Which of the following is/are correct ?

(A) A is a matrix such that $A^2 = A$. If $(I + A)^8 = I + \lambda A$ then $\lambda + 1$ is equal to 256.

(B) A is a matrix such that $A^2 = A$. If $(I + A)^8 = I + \lambda A$ then $\lambda + 1$ is equal to 255.

(C) If A, B, C are matrices such that $|A|_{3\times3} = 3$, $|B_{3\times3}| = -1$ & $|C_{3\times3}| = 2$ then $|2ABC| = -12$

(D) If A is an orthogonal matrix of order 2, then $|A| = \pm 1$.

2) For any 3 × 3 matrix M, let |M| denote the determinant of M. Let

$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ and $F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$

If Q is a nonsingular matrix of order 3 × 3, then which of the following statements is (are) **TRUE** ?

(A) $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

F = PEP and

(B) $|EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$

(C) $\left|(EF)^3\right| > |EF|^2$

(D) Sum of the diagonal entries of P^{-1} EP + F is equal to the sum of diagonal entries of E + P^{-1} FP

3) Consider a regular polygon having '9' sides, then which of the following statements is/are true ?

(A) Total number of triangles which can be formed by using the vertices of the polygon, so that it's exactly two sides are common to the sides of regular polygon are '9'.

(B) Total number of triangles that can be formed by using the vertices of regular polygon, whose none of the side is common to the sides of regular polygon are '30'.

(C) Total number of diagonals of regular polygon are 27.

(D) Total number of diagonals of regular polygon are 36.

SECTION-II

1) A person has 6 friends and during a certain vacation he met them during several dinners. He found that he dinned with all the 6 exactly on one day. with every 5 of them on 2 days, with every 4 of them on 3 days, with every 3 on 4 days; with every 2 on 5 days. Furthers every friend was present at 7 dinners and every friend was absent at 7 dinners. The number of dinner(s) he had alone is equal to

2) The maximum number of values of t for which the system of equations
(a + 2t)x + by + cz = 0
bx + (c + 2t)y + az = 0
cx + ay + (b + 2t) z = 0
has non-trivial solutions is

3) If $a^2 + b^2 + c^2 + ab + bc + ca \le 0 \ \forall$ a, b, c ∈ R, then find the value of the determinant

$$\begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}$$

4) Consider three matrices $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}, B\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$. Then the value of the sum

$$\text{tr}(A) + \text{tr}\left(\frac{ABC}{2}\right) + \text{tr}\left(\frac{A(BC)^2}{4}\right) + \text{tr}\left(\frac{A(BC)^3}{8}\right) + ... + \infty$$

is

5) Let Let A be a 2 × 2 symmetric matrix such that $A\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}3\\7\end{bmatrix}$ and the determinant of A be 1. If $A^{-1} = \alpha A + \beta I$, where I is an identity matrix of order 2 × 2, then $\alpha + \beta$ equals

6) If number of all possible five digits numbers using 1,2,3,4,5 such that sum of digits is 21, is a two digits number XY, then (X + Y) is equal to (Repetition is allowed)

SECTION-I (iii)

1)

List-I		List-II	
(P)	The total number of three digit numbers the sum of whose digits is even is	(1)	less than 500
(Q)	The total number of proper divisors of $N = 2^5.3^3.5^2.7$ is	(2)	greater than 400
(R)	Exponent of 11 in 2012! is	(3)	greater than 100
(S)	The total number of selections of atleast one letter from word ALLEN CAREER is	(4)	less than 300
		(5)	greater than 200

(A) P → 1,2,3,5;Q → 1,3,4;R → 1,3,4;S → 1,2,3,5

(B) P → 1,2,3;Q → 1,2,4;R → 1,2,3,5;S → 1,5

(C) P → 1,2,3,5;Q → 1,2,3,4;R → 2,3,5;S → 2,4

(D) P → 1,3;Q → 1,3,4;R → 1,2,3;S → 1,2

2) **Answer the following by appropriately matching the lists based on the information given in the paragraph.**
Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Let $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_i, B_i, C_i are co-factor of a_i, b_i, c_i respectively and $\Delta' = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ then $\frac{\Delta^2}{\Delta'}$ is	(1)	1
(Q)	If $x > 0$ such that $\left(\sqrt{5}\right)^{\log_5 x}\left(\sqrt{5}\right)^{\log_{5^2} x}\left(\sqrt{5}\right)^{\log_{5^4} x}\left(\sqrt{5}\right)^{\log_{5^8} x}\ldots\infty = 7$ then x is	(2)	2

(R)	Number of solution(s) of the equation $\sin x + \cos x - 1 = \sin 2x - 2$ is $[0, 3\pi]$ is	(3)	3
(S)	Minimum value of $\frac{\left(x^2+y^2\right)\left(y^2+z^2\right)\left(z^2+x^2\right)}{8x^2y^2z^2}$ is	(4)	4
		(5)	7

(A) P → 1;Q → 4;R → 2;S → 1

(B) P → 2;Q → 5;R → 3;S → 1

(C) P → 1;Q → 5;R → 3;S → 1

(D) P → 2;Q → 4;R → 3;S → 1

3) Let A be a matrix of order 3 × 3 such that |A| = 2 then

List-I		**List-II**	
(P)	The value of |Adj A| is	(1)	2
(Q)	The value of |Adj 2A| is	(2)	2^2
(R)	The value of |Adj (Adj A)| is	(3)	2^4
(S)	If Adj (Adj A) = αA then the value of α is	(4)	2^6
		(5)	2^8

(A) P → 2;Q → 4;R → 3;S → 1

(B) P → 3;Q → 5;R → 3;S → 1

(C) P → 3;Q → 5;R → 4;S → 1

(D) P → 2;Q → 5;R → 3;S → 1

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4
A.	B	C	C	A

SECTION-I (ii)

Q.	5	6	7
A.	A,C	A,C,D	A,B,C,D

SECTION-II

Q.	8	9	10	11	12	13
A.	1.20	14.00	200.00	10.00	0.50	1.00

SECTION-I (iii)

Q.	14	15	16
A.	A	C	D

PART 2 : CHEMISTRY

SECTION-I (i)

Q.	17	18	19	20
A.	A	A	B	D

SECTION-I (ii)

Q.	21	22	23
A.	A,D	A,B,D	A,B,D

SECTION-II

Q.	24	25	26	27	28	29
A.	4.00	6.00	18.00	3.00	3.00	3.00

SECTION-I (iii)

Q.	30	31	32
A.	A	C	A

PART 3 : MATHEMATICS

SECTION-I (i)

Q.	33	34	35	36
A.	**B**	**A**	**C**	**B**

SECTION-I (ii)

Q.	37	38	39
A.	**A,D**	**A,B,D**	**A,B,C**

SECTION-II

Q.	40	41	42	43	44	45
A.	**1.00**	**3.00**	**65.00**	**6.00**	**5.00**	**7.00**

SECTION-I (iii)

Q.	46	47	48
A.	**A**	**C**	**D**

SOLUTIONS

PART 1 : PHYSICS

1)

Let v be the apparent depth of bubble then by using

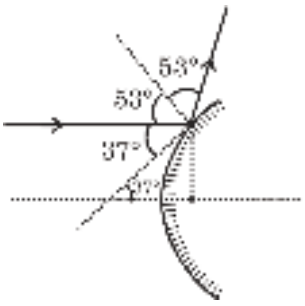
$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\Rightarrow \frac{1}{v} - \frac{1.5}{-4} = \frac{1-1.5}{-10}$$

$$\Rightarrow v = -\frac{40}{13}$$

v = - 3.1 cm

2)

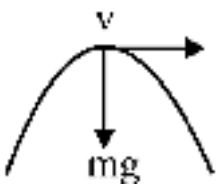


$$\tan\theta = \frac{dy}{dx} = \frac{3}{4}$$

$\theta = 37°$

Angle = 180 – 106 = 74°

3)



$N - 0$

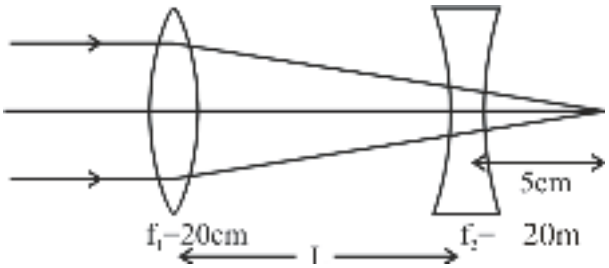
$$mg = \frac{mv^2}{a}$$

$$v = \sqrt{ag}$$

4)

Area under F – s curve gives the change in kinetic energy.

5)

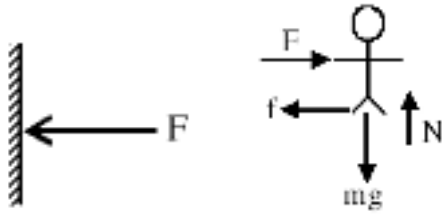


E_2 : $\frac{1}{v_2} - \frac{1}{u_2} = \frac{1}{f} \Rightarrow \frac{1}{5} - \frac{1}{u_2} = \frac{1}{-20}$

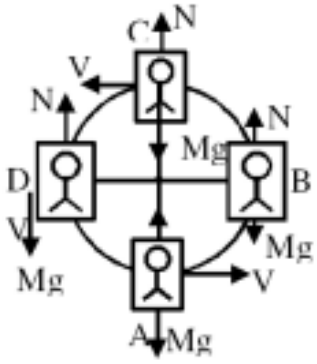
 u_2 = +4 cm
 L = 16 cm

6)

FBD of wall FBD of man



So men can apply a force
F = f – μmg
If μ > 1 then
f > mg i.e. force applied by men can be more than weight.
Due to action reach pair force will be same but opposite in direction.



7)
Weight reading at A

$$N - Mg = \frac{mV^2}{R}$$

$$N_A = Mg + \frac{MgR}{R} = 2Mg = 2w$$

At point C

$$Mg - N_C = \frac{mv^2}{R}$$

$N_C = mg - mg = 0$
So, weight reading at A is greater than the weight reading at C by 2w.

$$\text{At point B}$$
$$N_B = mg = W$$
$$\frac{N_A}{N_B} = \frac{2w}{w} = 2$$

8) Using $\mu = \frac{C}{V}$

$$\frac{\mu_2}{\mu_1} = \frac{v_1}{v_2} \Rightarrow \mu_2 = \left(\frac{v_1}{v_2}\right)\mu_1 = \left(\frac{2 \times 10^8}{2.5 \times 10^8}\right) \times 1.5$$

$\mu_2 = \frac{6}{5} \Rightarrow \mu_2 = 1.2$

9) $u = -\left(4 + \frac{4}{2}\right) \times 2 = -12$ cm

$\frac{1}{v} + \frac{1}{-12} = \frac{1}{-8}$

$\Rightarrow v = -24$ cm.

for final image (refraction through plane surface) we can write.

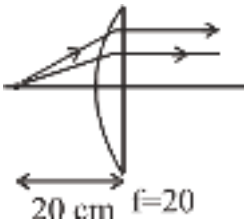
$24 = 2\left(\frac{x}{1} + \frac{4}{2}\right)$

$\Rightarrow 24 = 2x + 4$

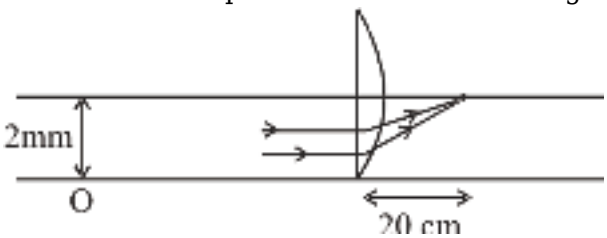
x = 10 cm

so distance from P is 14 cm.

10) Now lens 1 and lens 2 has f = 20 cm.



Now for lens 2 . p-axis is 2 mm above the given axis so image is at focus point



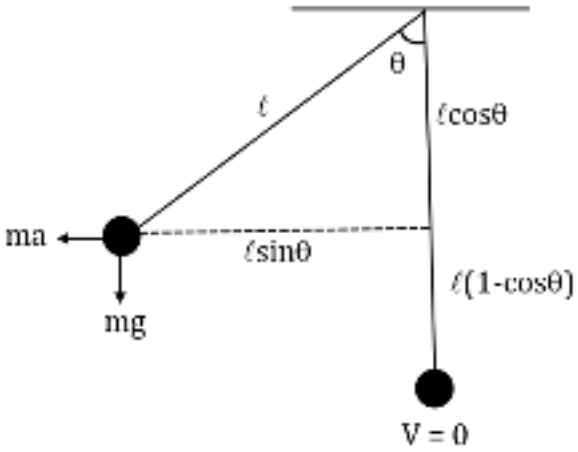
n = 40 cm ; m = 0.2 cm

so nm = 8

11) $w_f = \Delta k$

$\int_0^1 (0.1x) \times (0.1) \times 10dx = \frac{1}{2} \times 0.1 \left(0 - (0.1v)^2\right)$

= v = 10 m/s



12)

WET,

$W_g + W_P = \Delta k$

$-mg\ell(1-\cos\theta) + ma\ell\sin\theta = 0$

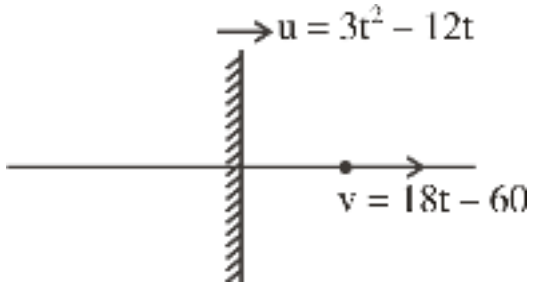
$g - g\cos\theta = a\sin\theta$

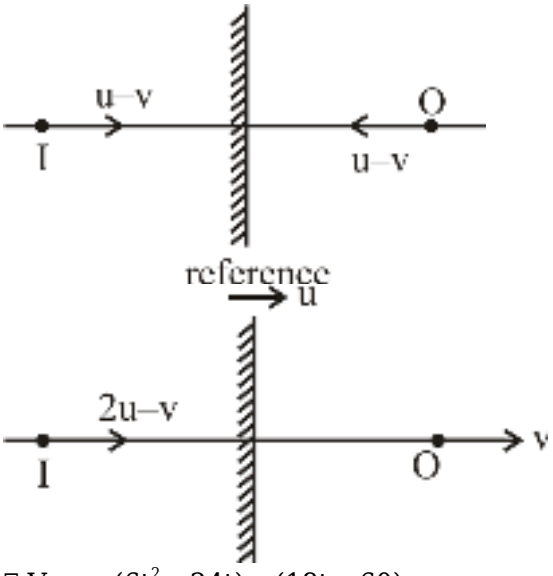
$\tan\frac{\theta}{2} = \frac{a}{g}$

13) $P = \frac{mgh}{T} = \frac{5 \times 60 \times 10 \times 20}{60} = 1000W$

[$\because m = 5 \times 60kg$; h = 20m and T = 60 sec]

14)





$\therefore V_{img} = (6t^2 - 24t) - (18t - 60)$

$= 6t^2 - 42t + 60$

(A) $0 = 6(t^2 - 7t + 10) = 6(t-2)(t-5)$

$\therefore$ t = 2 & 5

(B) a_{img} = 12t – 42

15) (A) Velocity of fish in air = $4 \times \frac{3}{4} = 3 \uparrow$
Velocity of fish w.r.t bird = 3 + 6 = $9 \uparrow$

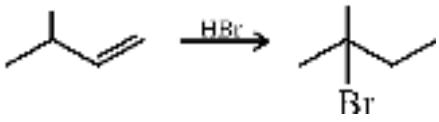
(B) Velocity of image of fish after reflection from mirror in air = $4 \times \frac{3}{4} = 3 \downarrow$
w.r.t bird = – 3 + 6 = $3 \uparrow$

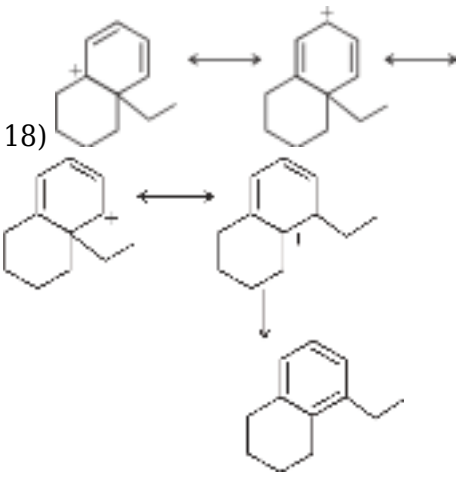
(C) Velocity of bird in water = $6 \times \frac{4}{3} = 8 \downarrow$
w.r.t fish = 8 + 4 = $12 \downarrow$

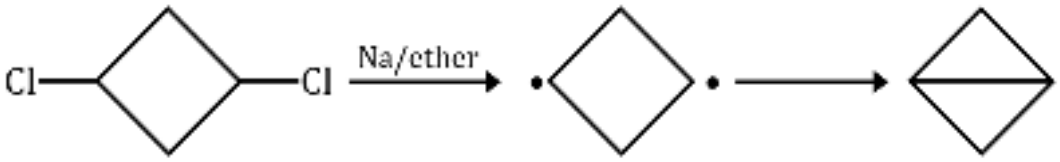
(D) Velocity of bird in water after reflection from mirror = $8 \uparrow$
w.r.t fish = 8 – 4 = $4 \uparrow$

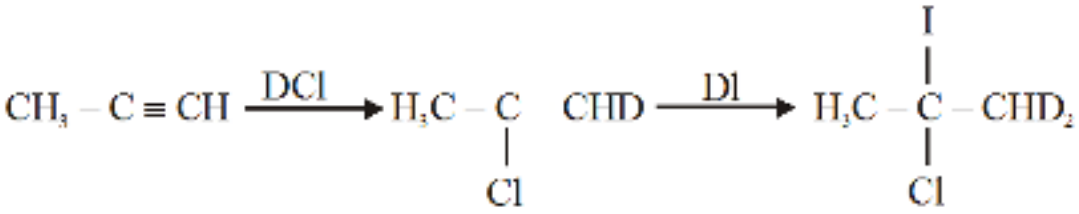
16) (A) → (P,T); (B) → (Q,R,S); (C) → (Q,T); (D) → (P,R,S)

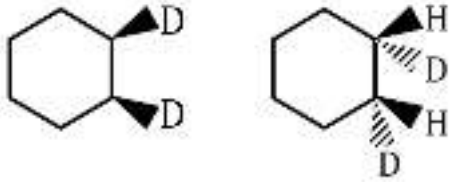
PART 2 : CHEMISTRY

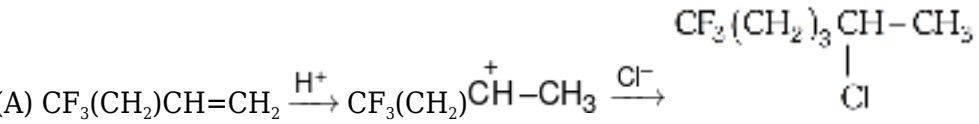
17) 

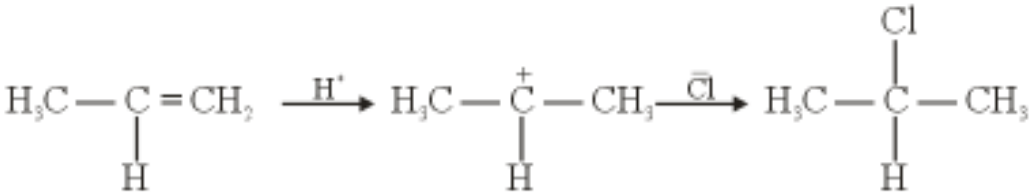
18) 

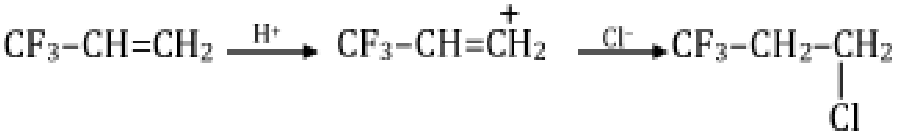
19) 

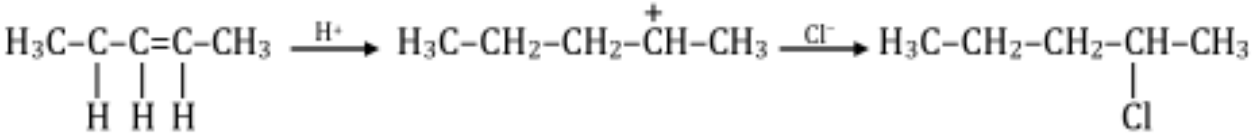
20) 

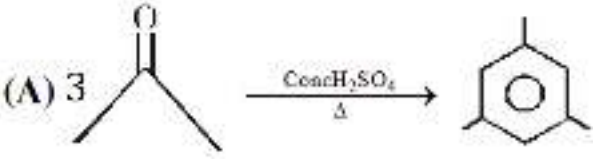
21) 

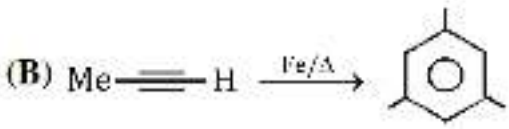
22) (A) 

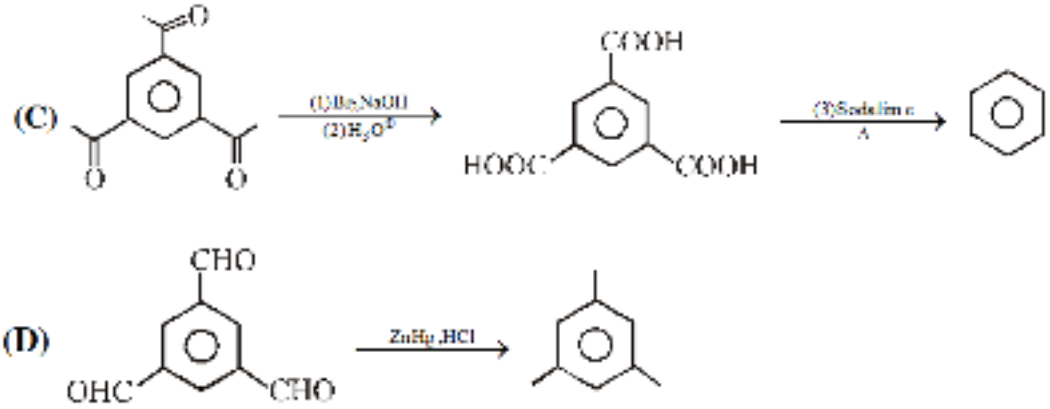
(B) 

(C) 

(D) 

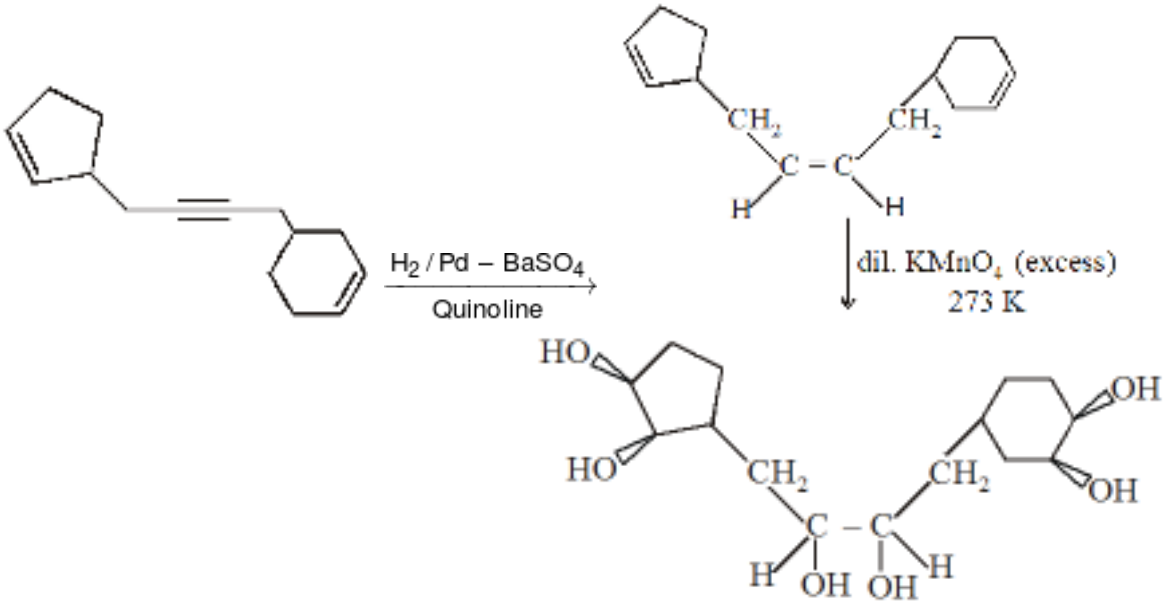
23) 





24) (i), (iv), (vi), (ix)

25)



total 6 –OH group present in a molecule of the major product.

26) $\underset{1\text{mole}}{C_3H_8} + \underset{5\text{mole}}{5O_2} \rightarrow 3CO_2 + 4H_2O$

For 1 mole propane combustion 5 mole O_2 required

$$C_4H_{10} + \frac{13}{2}O_2 \rightarrow 4CO_2 + 5H_2O$$

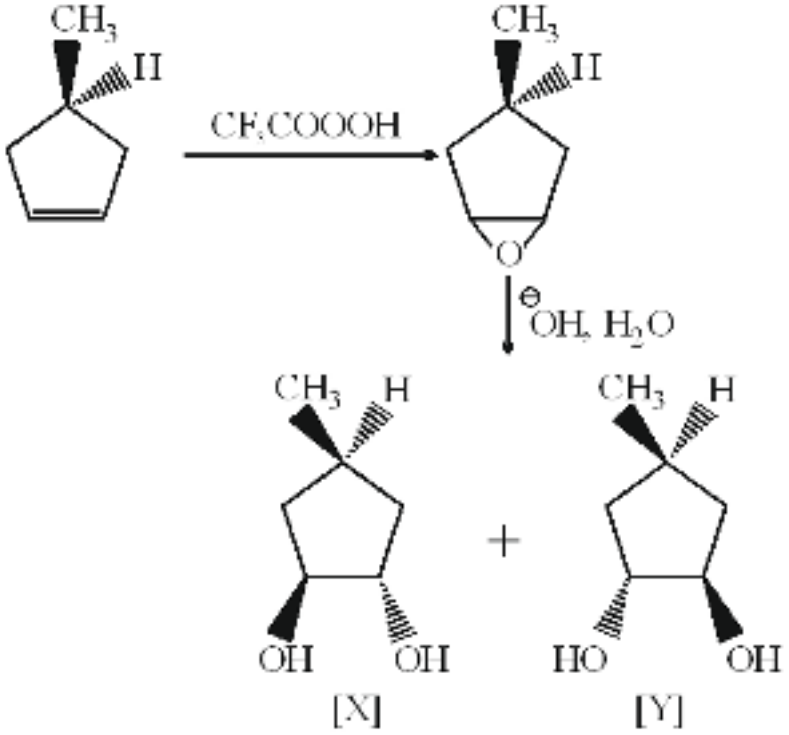
1 mole 6.5 mole

2 mole 13 mole

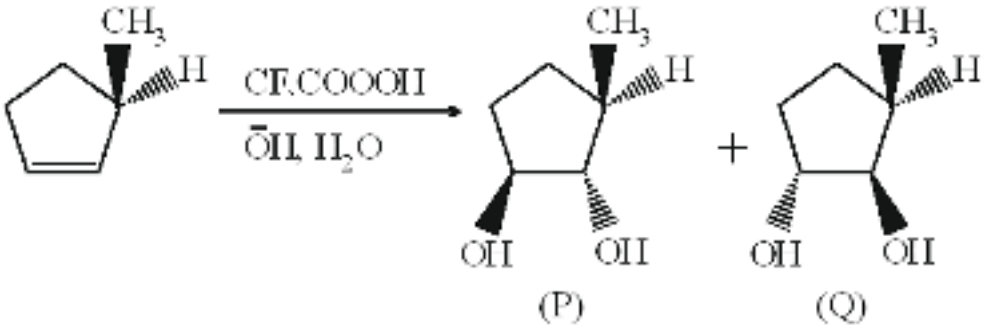
For 2 moles of butane 13 mole of O_2 is required

Total moles = 13 + 5 = 18

27) (i)

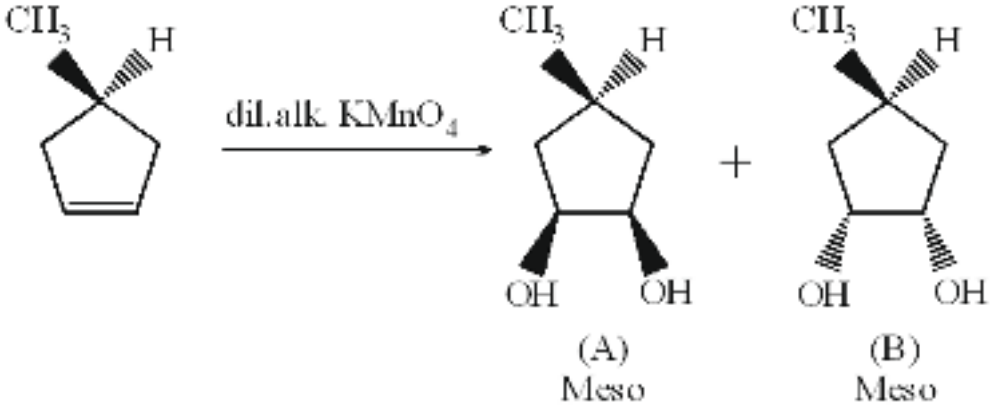


X and Y are enantiomer



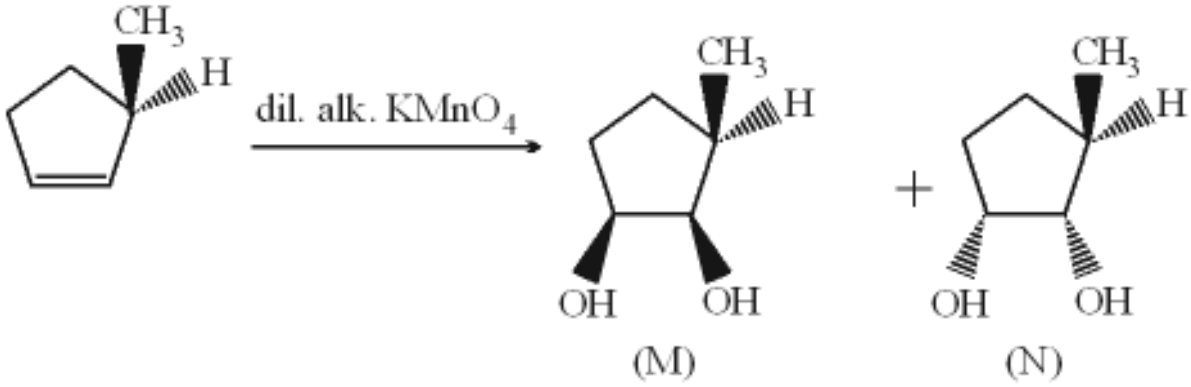
(ii)

P and Q are distereomer



(iii)

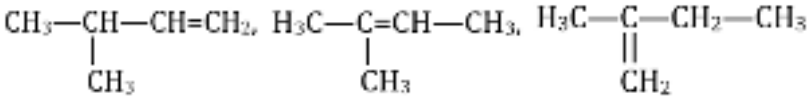
A and B are distereomer

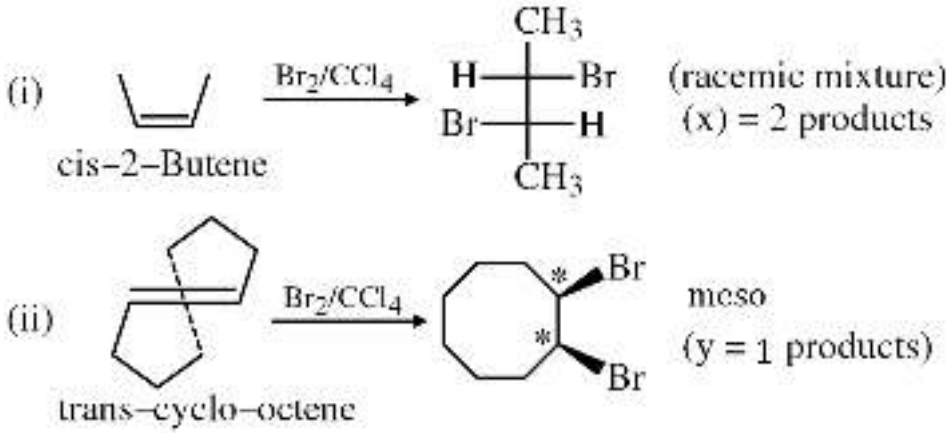


(iv)

Correct statements are (1), (3) & (6)

28) Number of possible alkene





29)

The value of (x + y) ⇒ 2 + 1

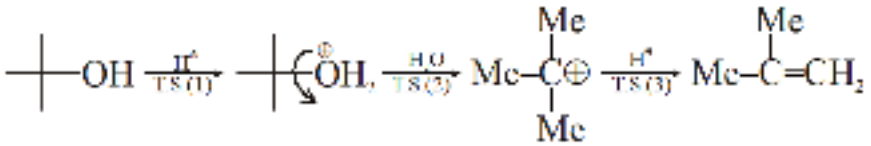
⇒ 3

30)

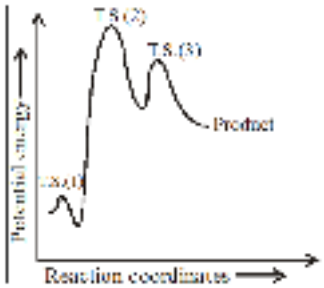
A. Reaction (a) is isomerisation reaction.
B. Reaction (b) is hydroxylation of 3° H atoms, which is carried out by alkaline $KMnO_4$. So, the answer is (p).
C. Reaction (c) is conversion of carbonyl group to (–CH_2) group, which is carried out by Clemmenson or Wolff-Kishner or HI + P reactions. So, the answer is (q, s, and u).
D. Reaction (d) is reduction of RX to RH, which is carried out by HI + Red P

31)

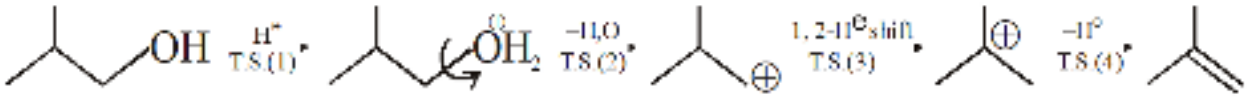
(A → R)



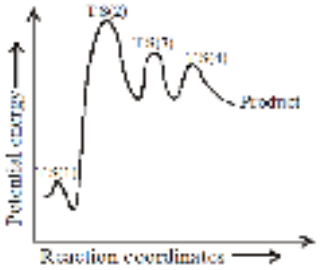
Number of transition state (T.S.) in this reaction is (3) So, energy curve of this reaction is (R)



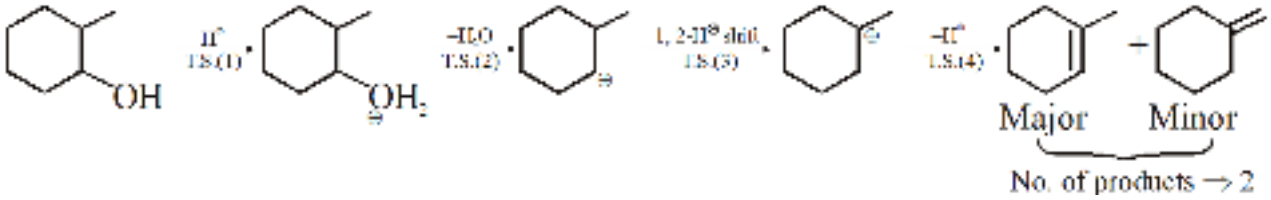
(B → T)



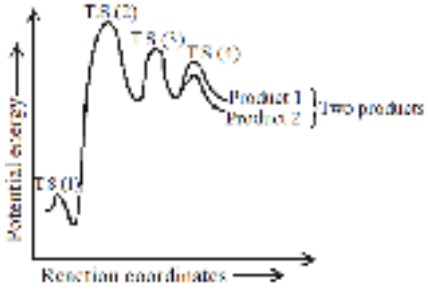
Number of T.S. in this reaction is (4) so energy curve is (T).



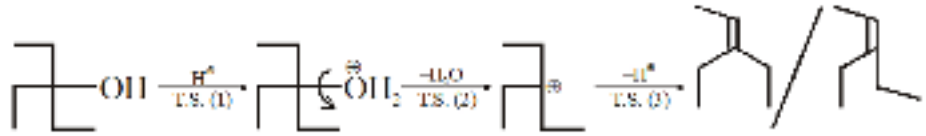
(C → P)



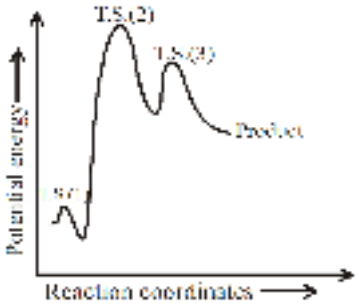
* No. of T.S. in this reaction = 4
No. of products in this reaction = 2
So, energy curve for this reaction is (P).

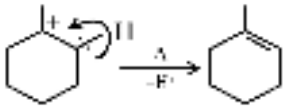


(D → R)



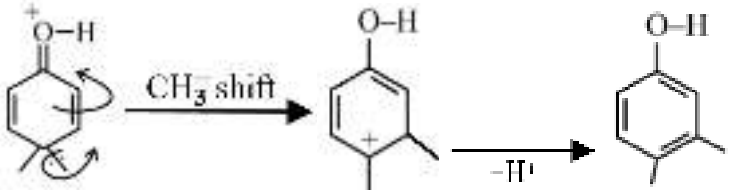
No. of T.S. in this reaction is = 3
No. of product in this reaction = 1
So, energy curve is (R)



32) (A) 

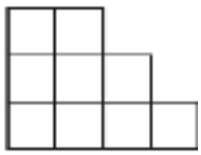
(B) $Me-C\equiv C-CH_2^+ \leftrightarrow CH_3-\overset{+}{C}=C=CH_2 \xrightarrow{H_2O}$

$H_3C-\overset{OH}{\overset{|}{C}}=C=CH_2 \rightleftharpoons H_3C-\overset{O}{\overset{||}{C}}-CH=CH_2$

(C) 

PART 3 : MATHEMATICS

33) **Case I :**
Two rows contain 2 'X' each and one row has 1 X .

Possible ways = 

$4 \times {}^3c_2 + {}^4c_2 \times {}^3c_1 + {}^4c_2 \times {}^3c_2 \times 2$

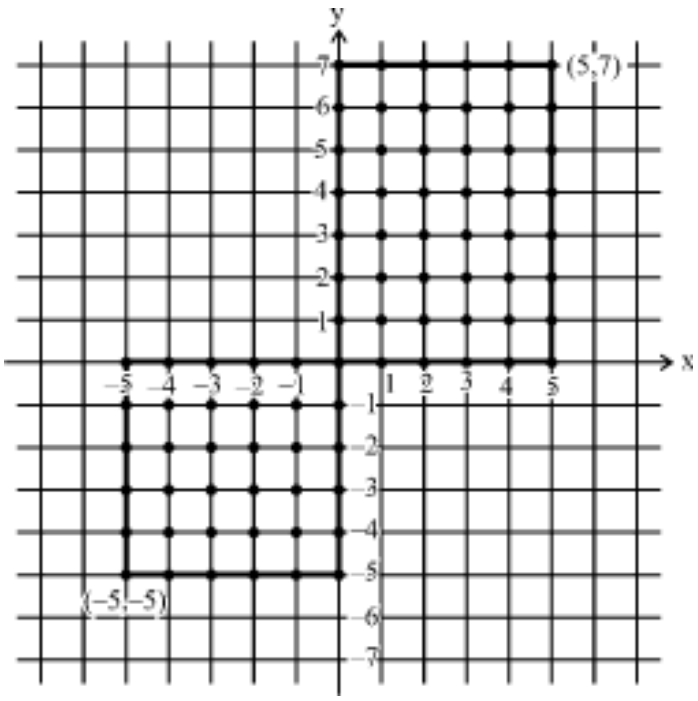
$= 12 + 18 + 36 = 66$

Case II :
One row has 3 'X' and two others have 1 X each

ways $= {}^4c_3 \times {}^3c_1 \times {}^2c_1 + {}^4c_1 \times {}^2c_1$

$= 4 \times 3 \times 2 + 4 \times 2 = 24 + 8 = 32$

Hence (B) is correct.

34) 

35) Determinant = $\begin{vmatrix} 0 & \sin\frac{\pi}{5}\sin\frac{2\pi}{5} & \tan\frac{667\pi}{4} \\ \sin\frac{3\pi}{5} & 0 & 0 \\ \tan\frac{3\pi}{10} & 0 & \sin\frac{4\pi}{5} \end{vmatrix}$

$$= -\sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\frac{3\pi}{5}\sin\frac{4\pi}{5}$$

$$= -\sin\frac{\pi}{5}\sin\frac{2\pi}{5}\sin\left(\pi-\frac{2\pi}{5}\right)\sin\left(\pi-\frac{\pi}{5}\right)$$

$$= -\frac{1}{4}\left[\cos\frac{\pi}{5}-\cos\frac{3\pi}{5}\right]^2 = -\frac{1}{4}\left[\frac{\sqrt{5}+1}{4}+\frac{\sqrt{5}-1}{4}\right]^2$$

$$= -\frac{1}{4}\times\frac{5}{4} = \frac{-5}{16}$$

36) $\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$

–bc – 2a(c – b) + a(4c – 3b) = 0
–bc – 2ac + 2ab + 4ac – 3ab = 0
2ac = ab + bc

37)

Conceptual

38) $PEP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

$\begin{pmatrix} 1 & 2 & 3 \\ 8 & 13 & 18 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{pmatrix}$

$P^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(B) $\left|EQ + PFQ^{-1}\right| = |EQ| + \left|PFQ^{-1}\right|$

$|E| = 0$ and $|F| = 0$ and $|Q| \neq 0$

$|EQ| = |E|\,|Q| = 0$, $\left|PFQ^{-1}\right| = \frac{|P|\,|F|}{|Q|} = 0$

$T = EQ + PFQ^{-1}$

$TQ = EQ^2 + PF = EQ^2 + P^2\,EP = EQ^2 + EP = E(Q^2 + P)$

$|TQ| = \left|E\left(Q^2 + P\right)\right| \Rightarrow |T|\,|Q| = |E|\left|Q^2 + P\right| = 0 \Rightarrow |T| = 0$ (as $|Q| \neq 0$)

(C) $\left|(EF)^3\right| > |EF|^2$

Here $0 > 0$ (false)

(D) as $P^2 = I \Rightarrow P^{-1} = P$ so $P^{-1}\,FP = PFP = PPEPP = E$

so $E + P^{-1})\,FP = E + E = 2E$

$P^{-1}\,EP + F \Rightarrow PEP + F = 2PEP$

$Tr(2PEP) = 2Tr(PEP) = 2Tr(EPP) = 2Tr(E)$

39) (A) Exactly two sides are common = 9 tringles

(B) Total no. of diagonals = $9c_2$–9 = 27

(C) Total no. triangles, so that none of side is common to the sides of regular polygon = $9c_3$ –9–9×5=84–9–45=30.

40) Let A_t, t = 1,2,... 6 be the set of days on which the friend is present at dinner and B_t be the set of days on which the friend is absent at dinner. Then $|A_t| = |B_t| = 7$

Also, $|A_i \cap A_j| = 5, |A_i \cap A_j \cap A_k| = 4$

$|A_i \cap A_j \cap A_k| \cap A_l| = 3$

$|A_i \cap A_j \cap A_k \cap A_l \cap A_m| = 2$ and

$|A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5 \cap A_6| = 1$

Where i, j, k, /, m, vary from 1 to 6 and are distinct. Now the number of dinners at which at least one friend was present

$= |A_1 \cup A_2 \cup .... \cup A_6|$

$= \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k|$

$- \sum |A_i \cap A_j \cap A_k \cap A_l|$

$+ \sum |A_i \cap A_j \cap A_k \cap A_l \cap A_m| - |A_1 \cap A_2 \cap ... \cap A_6|$

$= {}^6C_1 \times 7 - {}^6C_2 \times 5 + {}^6C_3 \times 4 - {}^6C_4 = 1$

$\times 3 + {}^6C_5 \times 2 - {}^6C_6 \times 1 = 13$

Total number of dinners is $|A_i| + |B_i|$

$= 7 + 7 = 14$

$\therefore$ Number of dinners the person has alone

$= 14 - 13 = 1$

41) $\Delta = \begin{vmatrix} a+2t & b & c \\ b & c+2t & a \\ c & a & b+2t \end{vmatrix} = 0$

Clearly Δ is a cubic polynomial in t and has 3 roots.

42)

$a^2 + b^2 + c^2 + ab + bc + ca \le 0$

$(a + b)^2 + (b + c)^2 + (c + a)^2 = 0$

$a + b = 0 \quad b + c = 0 \quad c + a = 0$

$a - b - c = 0$

$\begin{vmatrix} 4 & 0 & 1 \\ 1 & 4 & 0 \\ 0 & 1 & 4 \end{vmatrix}$

$4(16) + 1(1) = 65$

43) $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$

$BC = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$S = tr(A) + tr\left(\frac{A}{2}\right) + \tan\left(\frac{A}{4}\right) .....\infty$

$S = 3 + \frac{3}{2} + \frac{3}{4} .....\infty$

$S = \frac{3}{1 - \frac{1}{2}} = 6$

44) Let $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$

$\begin{bmatrix} a & b \\ b & d \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$, ad − $b^2 = 1$

a + b = 3, b + d = 7, (3 – b) (7 – b) − $b^2 = 1$

21 – 10b = 1 → b = 2, a = 1, d = 5

$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$

$A^{-1} = \alpha A + \beta I$

$\begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} \alpha+\beta & 2\alpha \\ 2\alpha & 5\alpha+\beta \end{bmatrix}$

α = −1, β = 6 → α + β = 5

45) 55551 ⇒ $\frac{5!}{4!} = 5$ numbers

55533 ⇒ $\frac{5!}{3!2!} = 10$ numbers

55542 ⇒ $\frac{5!}{3!} = 20$ numbers

55443 ⇒ $\frac{5!}{2!2!} = 30$ numbers

54444 ⇒ $\frac{5!}{4!} = 5$ numbers

Total = 70 numbers

46) (P) The number of three digits numbers

= 9 × 10 × 10

Half of which will have sum of digits even.

(Q) Total divisors of N

= (5 + 1)(3 + 1)(2 + 1)(1 + 1)

= 6.4.3.2 = 144

⇒ Number of proper divisors = 144 –2 = 142

(R) Exponent of 11 = $\left[\frac{2012}{11}\right] + \left[\frac{2012}{11^2}\right] + \left[\frac{2012}{11^3}\right]$ = 182 + 16 + 1 = 199

(S) 2A, 3E, 2L, 2R, N, C

Selections of atleast one alphabet can be done in (2 + 1)(3 + 1)(2 + 1)(2 + 1)2^2 – 1

= 3.4.3.3.4 – 1 = 431

47) (P) $\Delta` = \Delta^2 \Rightarrow \frac{\Delta^2}{\Delta'} = 1$

(Q) $\left(\sqrt{5}\right)^{\log_5 x\left(1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots\right)} = 7$

$\left(\sqrt{5}\right)^{(\log_5 x)\times 2} = 7$ ⇒ x = 7

(R) (sinx + cosx) – 1 = (sinx + cosx)^2 – 1 – 2

sinx + cosx = t ⇒ t^2 – t – 2 = 0

t = 2 or t = –1

sinx + cosx = 2 is rejected &

sinx + cosx = –1

⇒ sinx = –1 or cosx = –1

$x = \frac{3\pi}{2}$, x = π, 3π

total solutions are 3

(S) $\frac{x^2+y^2}{2} \geqslant xy$ (A.M. ≥ G.M.)

$\frac{\left(y^2+z^2\right)}{2} \geqslant yz$

$\frac{\left(z^2+x^2\right)}{2} \geqslant zx$

multiply all of them

48) (P) $|\text{Adj } A| = |A|^{n-1} = 22 = 4$

(Q) $|\text{Adj } 2A| = 2^{n^2-n}\,|A|^{n-1} = 2^6 \cdot 2^2$

$|\text{Adj } 2A| = 2^8$

(R) $|\text{Adj (Adj } A)| = |A|^{(n-1)^2} = 2^{2^2} = 2^4$

(S) $\text{Adj(Adj } A) = |A|^{n-2}.A = (2^{3-2}).A = 2A$